Linear Independence, $\vec{\S}$1.7

Recall from last class:

- A linear system is called **homogeneous** if it can be written as $A\vec{x} = \vec{0}$ where A is an $m \times n$ matrix, $\vec{x} \in \mathbb{R}^n$, and $\vec{0}$ is the vector of m zeros.
- **Theorem:** If $A\vec{x} = \vec{b}$ is consistent for a given $\vec{b}$, and $\vec{p}$ is a solution, then the solution set of $A\vec{x} = \vec{b}$ is the set of vectors of the form $\vec{p} + \vec{v}_h$ where $\vec{v}_h$ is a solution of $A\vec{x} = \vec{0}$.

Definition 1. *A set of vectors* $\{\vec{v}_1, \ldots, \vec{v}_p\}$ *in* $\mathbb{R}^n$ *is said to be* **linearly independent** *if*

$$x_1\vec{v}_1 + x_2\vec{v}_2 + \cdots + x_p\vec{v}_p = \vec{0}$$

has only the trivial solution. The set is called **linearly dependent** *if the equation has non-trivial solutions and the equation is called the* **dependence relation**.

1. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \\ 3 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 4 \\ 5 \\ 6 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix}$. Is $\{\vec{v}_1, \vec{v}_2, \vec{v}_3\}$ linearly independent?

Let $A = [\vec{a}_1 \ \ldots \ \vec{a}_n]$, then $\{\vec{a}_1 \ \ldots \ \vec{a}_n\}$ are linearly independent if and only if $A\vec{x} = \vec{0}$ has only the trivial solution.

2. Is a set with only one vector always linearly independent?

3. If $\vec{0} \in \{\vec{v}_1, \ldots, \vec{v}_p\}$, then is the set linearly dependent?

4. If $\vec{0} \in \text{Span}\{\vec{v}_1, \ldots, \vec{v}_p\}$, then the vectors are linearly independent?

5. Let $\vec{v}_1 = \begin{bmatrix} 1 \\ 2 \end{bmatrix}$, $\vec{v}_2 = \begin{bmatrix} 3 \\ 4 \end{bmatrix}$, and $\vec{v}_3 = \begin{bmatrix} 5 \\ 6 \end{bmatrix}$. Are the vectors linearly independent?

Theorem 7: A set $S = \{\vec{v}_1, \ldots, \vec{v}_p\}$ of two or more vectors is linearly dependent if and only if one of the vectors of S is a linearly combination of the others.

Moreover if $\vec{v}_1 \neq \vec{0}$, then some $\vec{v}_j$ for $j > 1$ is a linear combination of $\vec{v}_1, \ldots, \vec{v}_{j-1}$.

6. Prove Theorem 7.

7. Let $\vec{u} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$ and $\vec{v} = \begin{bmatrix} 2 \\ 0 \\ 8 \end{bmatrix}$

(a) Describe $\text{Span}\{\vec{u}, \vec{v}\}$

(b) Prove $\vec{w} \in \text{Span}\{\vec{u}, \vec{v}\}$ if and only if $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent.

Theorem 8: Any set $\{\vec{v}_1, \ldots, \vec{v}_p\}$ in $\mathbb{R}^n$ is linearly dependent if $p > n$.

8. Prove the theorem.

Linear Independence – Answers / Solutions

1. In order to determine if the vectors are linearly independent, we need to solve the vector equation $c_1\vec{v}_1 + c_2\vec{v}_2 + c_3\vec{v}_3 = \vec{0}$. We instead solve the corresponding augmented matrix. Row-reducing the augmented matrix, we get

$$\left(\begin{array}{ccc|c} 1 & 4 & 2 & 0 \\ 2 & 5 & 1 & 0 \\ 3 & 6 & 0 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & -2 & 0 \\ 0 & 1 & 1 & 0 \\ 0 & 0 & 0 & 0 \end{array}\right).$$

Thus we get $c_1 - 2c_3 = 0$ and $c_2 + c_3 = 0$, and hence the solution set is given by

$$\left\{\vec{c} \in \mathbb{R}^3 : c_1 = 2c_3, c_2 = -c_3\right\} = \left\{\begin{bmatrix} 2c_3 \\ -c_3 \\ c_3 \end{bmatrix} : c_3 \in \mathbb{R}\right\} = \left\{c_3\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix} : c_3 \in \mathbb{R}\right\} = \text{Span}\left\{\begin{bmatrix} 2 \\ -1 \\ 1 \end{bmatrix}\right\}.$$

Thus the vector equation has non-trivial solutions, and the vectors are linearly dependent. From the solution set we can see how the vectors are related to one another using a dependence relation. In this case, a dependence relation is

$$2\vec{v}_1 - \vec{v}_2 + \vec{v}_3 = \vec{0}.$$

2. No. Since the vector equation $c \cdot \vec{0} = \vec{0}$ has non-trivial solutions, we get $\vec{0}$ is linearly dependent as a single vector. On the other hand, if $\vec{v}$ is any non-zero vector, then it is linearly independent.

3. Yes. We prove the claim. Since $\vec{0} \in \{\vec{v}_1, \dots, \vec{v}_p\}$, we know $\vec{v}_k = \vec{0}$ for some k. Thus

$$0 \cdot \vec{v}_1 + \cdots + 0 \cdot \vec{v}_{k-1} + 1 \cdot \vec{v}_k + 0 \cdot \vec{v}_{k+1} + \cdots + 0 \cdot \vec{v}_p = \vec{0},$$

which gives a non-trivial solution to $x_1\vec{v}_1 + \cdots + x_n\vec{v}_p = \vec{0}$. Thus the vectors are linearly dependent.

4. No. $\vec{v} \in \text{Span}\{\vec{v}_1, \dots, \vec{v}_p\}$ whether or not the vectors are linearly independent.

5. Form the augmented matrix and row-reduce to get

$$\left(\begin{array}{ccc|c} 1 & 3 & 5 & 0 \\ 2 & 4 & 6 & 0 \end{array}\right) \sim \left(\begin{array}{ccc|c} 1 & 0 & -1 & 0 \\ 0 & 1 & 2 & 0 \end{array}\right).$$

Thus the vector equation has non-trivial solutions, and the vectors are linearly dependent.

6. *Proof.* ($\Longrightarrow$). Suppose S is linearly dependent. By definition of linear dependence, there exists $c_1, \dots, c_p$, not all zero, such that

$$c_1\vec{v}_1 + \cdots + c_p\vec{v}_p = \vec{0}.$$

Since not all c_k are zero, we know there exists some $c_k \neq 0$. We solve the vector equation for $\vec{v}_k$ to get

$$\vec{v}_k = -\frac{c_1}{c_k}\vec{v}_1 - \cdots - -\frac{c_{k-1}}{c_k}\vec{v}_{k-1} - \frac{c_{k+1}}{c_k}\vec{v}_{k+1} - \cdots - \frac{c_p}{c_k}\vec{v}_p.$$

Thus $\vec{v}_k$ can be written as a linear combination of the other vectors.

($\Longleftarrow$). Suppose one of the vectors can be written as a linear combination of the others. We can assume $\vec{v}_p$ can be written in terms of the others (if not we can reorder the vectors). Thus there exists $c_1, \dots, c_{p-1}$ such that

$$\vec{v}_p = c_1\vec{v}_1 + \cdots + c_{p-1}\vec{v}_{p-1}.$$

Rearranging we get

$$\vec{0} = c_1\vec{v}_1 + \cdots + c_{p-1}\vec{v}_{p-1} - \vec{v}_p.$$

Thus we have a non-trivial solution to $x_1\vec{v}_1 + \cdots + x_p\vec{v}_p = \vec{0}$, and thus the vectors are linearly dependent. $\square$

7.

(a) In this problem we have $\text{Span}\{\vec{u}, \vec{v}\}$ is the xz-plane in $\mathbb{R}^3$.

(b) *Proof.* ($\Longrightarrow$). Suppose $\vec{w} \in \text{Span}\{\vec{u}, \vec{v}\}$, then $\vec{w}$ can be written as a linear combination of $\vec{u}$ and $\vec{v}$. Thus, by Theorem 7, we get $\{\vec{u}, \vec{v}, \vec{w}\}$ is linearly dependent.

($\Longleftarrow$). Since $\vec{u} \neq 0$ and $\vec{v}$ is not parallel with $\vec{u}$, we can use Theorem 7 to conclude that $\vec{w}$ is a linear combination of $\vec{u}$ and $\vec{v}$, and hence $\vec{w}$ is in the span of $\vec{u}$ and $\vec{v}$. $\square$

8. *Proof.* Let $A = [\vec{v}_1 \ \dots \ \vec{v}_p]$ with $\vec{v} \in \mathbb{R}^n$ and $p > n$. Thus $A\vec{x} = \vec{0}$ gives a linear system of equations with n equations and p variables. Thus we can have at most n pivots. Since $p > n$, we have at least one non-pivot column, and hence we have a free variable. Thus $A\vec{x} = \vec{0}$ has non-trivial solutions, and thus the columns are linearly dependent. $\square$